NEPAL COLLEGE OF INFORMATION TECHNOLOGY

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A

Project Report on

Effect of Different Types of Water

On

Strength of Hardened Concrete

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ABSTRACT

Concrete is the most widely used construction material globally, and its performance is critically dependent on the quality of its constituent materials. While significant attention is given to cement and aggregates, the quality of mixing water is often overlooked, despite its crucial role in the cement hydration process which governs strength and durability. This project presents an experimental investigation into the effects of different types of water on the key properties of hardened concrete.

The primary objective of this study was to compare the compressive strength and workability of concrete prepared with three distinct water sources: purified jar water, standard tap water, and salty water. A concrete mix of a specified grade was designed as per IS 10262:2019, maintaining a constant water-cement ratio across all batches. The workability of the fresh concrete mix for each water type was evaluated using the slump cone test (IS 1199). For strength analysis, 150mm x 150mm x 150mm concrete cubes were cast, cured in water, and tested for compressive strength at 7 days using a Compression Testing Machine (CTM) as per IS 516:2018.

The results revealed that concrete produced with jar water and tap water showed similar and satisfactory performance in both workability and compressive strength, achieving the desired design strength. In contrast, the concrete mixed with salty water exhibited a lower slump value and a significant reduction in compressive strength, particularly at the 7-day mark, compared to the other two mixes.

This study concludes that the presence of dissolved salts and impurities in mixing water has a detrimental effect on the ultimate strength of concrete. The findings underscore the critical importance of using clean, potable water in concrete production to ensure structural integrity, and validate the cautionary guidelines specified in construction codes like IS 456.

Keywords: *Concrete Strength, Water Quality, Compressive Strength, Workability, Salty Water, Cement Hydration.*

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LIST OF ABBREVIATIONS

The following abbreviations and symbols have been used in this report:

General Abbreviations

Abbreviation	Full Form
C-S-H	Calcium Silicate Hydrate (gel)
CCN	Cement Chemist Notation
CH	Calcium Hydroxide (Portlandite)
CTM	Compression Testing Machine
C_2S	Dicalcium Silicate (Belite)
C_3S	Tricalcium Silicate (Alite)
IS	Indian Standard
JW	Jar Water (Project-specific identifier)
M25	Mix 25 (Concrete grade of 25 N/mm^2)
NaCl	Sodium Chloride (Common Salt)
OPC	Ordinary Portland Cement
SSD	Saturated Surface Dry
SW	Salty Water (Project-specific identifier)
TW	Tap Water (Project-specific identifier)
w/c	Water-Cement Ratio

Units of Measurement

Symbol	Description
°C	Degrees Celsius
kg	Kilogram
kg/m^3	Kilograms per cubic meter
KN	KiloNewton
m^3	Cubic meter
mm	Millimeter
MPa	Megapascal (equivalent to N/mm^2)
N/mm^2	Newtons per square millimeter

CHAPTER-1: INTRODUCTION

1.1 Background

1.1.1 Basics of Concrete and its Strength

Concrete is a composite material that has become the most widely used construction material on Earth due to its strength, versatility, durability, and cost-effectiveness. It is fundamentally composed of four main ingredients: cement, fine aggregate (such as sand), coarse aggregate (such as gravel or crushed stone), and water. These components are mixed in specific proportions to form a workable paste that can be cast into any desired shape.

The strength of concrete is not inherent but is developed through a chemical reaction known as hydration. The chemical process of hydration is highly complex, but the primary strength-giving reactions involve the hydration of the two main silicate compounds in Portland cement: Tricalcium Silicate (C_3S) and Dicalcium Silicate (C_2S). These reactions are best represented using Cement Chemist Notation (where C = CaO, S = SiO_2, H = H_2O).

The two principal reactions are:

1. Hydration of Tricalcium Silicate (Alite - C_3S): This is a rapid reaction responsible for the initial set and most of the early-age strength (first 7 to 28 days).

$2(3CaO{\cdot}SiO_2) + 6H_2O \rightarrow 3CaO{\cdot}2SiO_2{\cdot}3H_2O + 3Ca(OH)_2$

In Cement Chemist Notation:

$2C_3S + 6H \rightarrow C_3S_2H_3 + 3CH$

$C_3S_2H_3$ represents Calcium Silicate Hydrate (C-S-H gel), the primary binding agent in concrete. It is an amorphous gel that forms a solid matrix and is the main source of concrete's strength.

CH represents Calcium Hydroxide (Portlandite), a crystalline byproduct that contributes to strength but can be susceptible to chemical attack.

2. Hydration of Dicalcium Silicate (Belite - C_2S): This is a much slower reaction that contributes primarily to the long-term strength gain of concrete (beyond 28 days).

$2(2CaO{\cdot}SiO_2) + 4H_2O \rightarrow 3CaO{\cdot}2SiO_2{\cdot}3H_2O + Ca(OH)_2$

In Cement Chemist Notation:

$2C_2S + 4H \rightarrow C_3S_2H_3 + CH$

Together, these reactions consume the original cement particles and water to form the strong, solid, and durable cement paste.

This paste acts as a binder, coating the surfaces of the aggregates and gluing them together into a solid mass. The aggregates provide bulk and volume to the concrete, acting as an inert filler, while the hydrated cement paste provides the strength. The strength development is a gradual process, with concrete gaining most of its design strength within the first 28 days of casting, though it continues to harden for months or even years.

1.1.2 Importance of Compressive Strength and Durability

Among the various properties of hardened concrete, compressive strength is universally considered the most important. It is a measure of the material's ability to resist axial loads or squeezing forces that tend to crush it. Structural elements like columns, foundations, and dams are designed primarily to resist compressive forces, making this property a critical parameter for ensuring the safety and stability of a structure. The compressive strength of concrete is the principal measure used for quality control on a construction site and for verifying that the concrete meets the specifications outlined by the structural or civil engineer (e.g., M20, M25 grade).

While strength is essential for immediate structural performance, durability is critical for long-term performance. Durability is defined as the ability of concrete to resist weathering, chemical attack, abrasion, and other deterioration processes throughout its intended service life. A structure that is strong but not durable will fail prematurely, leading to costly repairs and potential safety hazards. The durability of concrete is intrinsically linked to its density and porosity. A well-hydrated, dense concrete with low permeability is less susceptible to the ingress of harmful substances like chlorides, sulfates, and acids, which can degrade the cement matrix and corrode the embedded steel reinforcement. Therefore, both high initial strength and long-term durability are essential for creating safe, reliable, and sustainable infrastructure.

1.2 Problem Statement

The final performance of hardened concrete is highly sensitive to the proportions and quality of its constituent materials. In modern construction, factors such as the water-cement (w/c) ratio, aggregate grading, and cement type are meticulously controlled because their effects on strength and durability are well-documented and understood.

However, amidst this rigorous control, the quality of the mixing water itself is often treated as a secondary consideration or simply assumed to be adequate, provided it appears clean. This assumption can be a critical oversight. Water in concrete is not merely a lubricant for the mix; it is an active chemical reactant in the hydration process. The presence of impurities—such as dissolved salts (chlorides, sulfates), suspended solids, organic matter, or industrial wastes—can significantly interfere with this chemical reaction. These contaminants can alter the setting time, reduce the ultimate strength, and compromise the long-term durability of the concrete, potentially leading to premature structural failure.

In practical field scenarios, the available water source may vary from treated potable water (tap water) and commercially purified water (jar water) to questionable sources like saline water in coastal areas. While standards like IS 456-2000 provide general guidelines on permissible limits for impurities, there is often a lack of direct, comparative experimental data readily available to site engineers on how these common water types specifically affect fundamental concrete properties like workability (slump) and compressive strength. This knowledge gap can lead to inconsistent construction quality and unforeseen risks to the service life of a structure.

1.3 Objectives

The primary aim of this project is to conduct a systematic study on how different types of commonly available water affect the fundamental properties of concrete. To achieve this, the following specific objectives have been established:

- To investigate and compare the workability of fresh concrete mixes prepared using purified jar water, standard tap water, and saline (salty) water, as measured by the slump test.
- To determine and compare the compressive strength of hardened concrete cubes made with each of the three water types at the curing ages of 7 days and 28 days.
- To analyze the collected data to draw conclusions about the suitability of each water type for use in concrete production and to quantify the potential impact of saline water on concrete strength.

1.4 Scope

The scope of this project is precisely defined by the materials and test methods employed. The study encompasses the following:

- Water Types: The investigation focuses on three specific water sources for creating concrete mixes: purified jar water, standard tap water, and prepared salty water.
- Concrete Mix: A single mix design, maintaining a constant water-cement ratio, cement quantity, and aggregate type, was used across all tests to ensure a direct comparison.
- Properties Investigated: The study evaluates two key properties of concrete:
    1. The workability of fresh concrete, measured by the slump test.
    2. The compressive strength of hardened concrete, measured at 7 and 28 days.
- Specimens: The experimental program involves the casting and testing of 150mm x 150mm x 150mm concrete cube specimens.

1.5 Significance and Limitations

Significance:

This study provides practical and valuable insights for the construction industry. By directly comparing the effects of common water types, the findings of this project can help site engineers and project managers make more informed decisions regarding material selection. It reinforces the importance of water quality as a critical parameter in concrete quality control, potentially preventing premature deterioration and enhancing the long-term performance of structures. The results serve as a clear, experimental validation of the guidelines specified in national standards like IS 456, highlighting the tangible risks associated with using substandard water.

Limitations:

While this research provides clear findings, it is important to acknowledge its limitations, which define the boundaries within which the conclusions are valid:

- Controlled Laboratory Conditions: The entire experiment was conducted in a controlled laboratory environment. This does not account for the wide variations in temperature, humidity, and curing practices that are common on actual construction sites, which can influence concrete properties.
- Limited Scope of Tests: The project focuses solely on slump and compressive strength. Other critical properties, such as flexural strength, tensile strength, and long-term durability parameters (e.g., water permeability, resistance to chemical attack), were not evaluated.
- Specific Water Contaminants: The study only investigates the effect of dissolved salts (specifically NaCl in the salty water). The results may not be generalizable to water containing other types of harmful contaminants, such as sulfates, alkalis, or industrial effluents.

CHAPTER 2: LITERATURE REVIEW

A thorough review of existing literature is essential to establish the theoretical framework for this study. This chapter summarizes the established knowledge regarding the role of water in concrete, the effects of its quality on concrete properties, and highlights the research gap that this project aims to address.

- ❖ The Fundamental Role of Water in Concrete: Neville (2011)

Water is an indispensable ingredient in concrete, playing a dual role. Firstly, it provides the necessary workability to the fresh mix, lubricating the aggregates and cement particles so the concrete can be properly mixed, transported, placed, and compacted. Secondly, and more importantly, water is a key chemical reactant in the hydration of cement, the process that transforms the plastic mix into a hard, solid mass. As Neville (2011) explains, the reaction between water and the silicate compounds in cement forms Calcium Silicate Hydrate (C-S-H) gel, which is the primary source of strength in concrete. The amount of water used, typically defined by the water-cement (w/c) ratio, is the single most important factor governing the strength and durability of concrete; a lower w/c ratio generally leads to higher strength and better durability.

- ❖ Water Quality Standards and Specifications: IS 456:2000

Recognizing the critical role of water, various national and international standards provide guidelines for its quality. IS 456:2000 (Plain and Reinforced Concrete - Code of Practice) stipulates that water used for mixing and curing concrete shall be clean and free from injurious amounts of oils, acids, alkalis, salts, sugar, organic materials, or other substances that may be deleterious to concrete or steel. As a general rule, the code suggests that water fit for drinking (potable water) is suitable for use in concrete. It also provides permissible limits for solids and certain chemical compounds like chlorides and sulfates, as their presence in excess can be harmful.

- ❖ Impact of Water Impurities on Concrete Properties: Mehta & Monteiro (2014), Gambhir (2013), Neville (2011)

Previous research has extensively documented the negative effects of impurities in mixing water.

- Effect on Compressive Strength: Mehta & Monteiro (2014) provide a detailed analysis of how various impurities interfere with the hydration process. Chlorides, commonly found in saline water or seawater, are particularly detrimental. While they can act as an accelerator and slightly increase early-age strength, they interfere with the long-term development of the C-S-H gel structure. This results in a more porous and weaker microstructure, leading to a significant reduction in the ultimate 28-day compressive strength. Sulfates, another common impurity, can react with the calcium aluminate in cement to form ettringite, an expansive compound that can cause internal cracking and degradation of the concrete matrix over time.

- Effect on Workability and Setting Time: The presence of dissolved salts can alter the surface tension of water and the flocculation of cement particles, thereby affecting the workability (slump) of the fresh mix. As noted by Gambhir (2013), high concentrations of salts can sometimes cause a "false set" or significantly accelerate the setting time, making the concrete difficult to place and finish properly.
- Effect on Durability and Reinforcement Corrosion: The most severe long-term effect of using saline water, particularly in reinforced concrete, is the risk of corrosion. Chlorides penetrate the hardened concrete and, upon reaching the steel reinforcement, break down its passive protective layer, initiating a corrosive electrochemical process. This leads to the formation of rust, which has a larger volume than the original steel, inducing internal pressure, cracking, and spalling of the concrete cover (Neville, 2011). This severely compromises the structural integrity and service life of the element.

❖ Research Gap:

While the literature provides extensive evidence on the harmful effects of specific chemical impurities like chlorides and sulfates, much of this research is focused on their chemical mechanisms or on the use of seawater under specific conditions. There is a practical gap in direct, comparative studies that evaluate the performance of concrete made with different types of commonly available water sources—such as commercially purified water, municipal tap water, and prepared salty water—all under the same experimental conditions.

This project aims to fill this gap by providing clear, quantifiable data on the impact of these three water types on workability and compressive strength. The findings will serve as a practical demonstration and validation of the established theories for students, educators, and construction professionals.

CHAPTER-3: MATERIALS AND METHODS

This chapter details the constituent materials, mix design procedure, and experimental methodologies employed to conduct the investigation. All procedures were carried out in accordance with relevant Indian Standards (IS) to ensure accuracy and consistency.

3.1 Materials Used

The selection and preparation of materials are fundamental to the validity of the experimental results. The following materials were used in this study.

3.1.1 Cement

Ordinary Portland Cement (OPC) of 43 Grade, conforming to the requirements of IS 269:2015, was used for all concrete mixes. The cement was procured in one batch and stored in an airtight container to prevent pre-hydration and maintain uniformity throughout the project. The cement was ensured to be fresh and free from any lumps before use.

3.1.2 Fine Aggregate

Locally sourced, clean river sand was used as the fine aggregate. The sand conformed to Zone II as per the grading requirements of IS 383:2016 (Specification for Coarse and Fine Aggregates from Natural Sources for Concrete). It was free from deleterious materials such as silt, clay, and organic impurities. Before mixing, the sand was air-dried to a saturated surface dry (SSD) condition.

3.1.3 Coarse Aggregate

Crushed angular stone with a nominal size of 20 mm was used as the coarse aggregate. The aggregate was hard, durable, and clean, conforming to the specifications of IS 383:2016. It was washed to remove dust and other fine particles and then air-dried to a saturated surface dry (SSD) condition prior to being used in the mix.

3.1.4 Water

To study the effect of water quality on concrete properties, three different types of water were used for mixing and curing, as detailed below. The baseline quality for mixing water was considered as per the guidelines of IS 456:2000.

- Jar Water (JW): Commercially available purified drinking water. This was used as a control sample representing water with minimal impurities and dissolved solids.
- Tap Water (TW): Standard potable water obtained directly from the municipal supply at the university laboratory. This represents the most commonly used water in urban construction projects.
- Salty Water (SW): This was prepared in the laboratory to simulate saline conditions. It was formulated by dissolving 35 grams of Sodium Chloride (NaCl) in 1 liter of tap

water, resulting in a salt concentration of approximately 3.5%. This concentration is commonly used to approximate the salinity of seawater.

3.2 Mix Design

The concrete mix was designed to achieve a target characteristic compressive strength of 25 N/mm² at 28 days, corresponding to M25 grade concrete. The mix proportioning was carried out in strict accordance with the guidelines specified in IS 10262:2019 - Concrete Mix Proportioning — Guidelines (Second Revision).

The primary objective of the mix design was to establish a single, consistent set of proportions for cement, fine aggregate, and coarse aggregate. This ensures that any observed differences in the workability and strength of the concrete can be directly attributed to the different types of water used.

The design was based on the following stipulations and material properties:

Grade Designation: M25

Cement Type: OPC 43 Grade

Maximum Nominal Size of Aggregate: 20 mm

Target Slump: 75 mm - 100 mm (for good workability)

Exposure Condition: Mild (as per IS 456:2000)

Water-Cement Ratio (w/c): A w/c ratio of 0.50 was selected based on the target strength and durability requirements.

Specific Gravity of Materials:

Cement: 3.15

Fine Aggregate: 2.65

Coarse Aggregate: 2.70

Based on the calculations as per IS 10262:2019, the final mix proportions of concrete were determined as shown in Table 3.1.

Table 3.1: Final Concrete Mix Proportions

Material	Quantity (kg)
Cement	…
Water	…
Fine Aggregate (Sand)	…
Coarse Aggregate (20mm)	…

Water-Cement Ratio = 0.50

The final mix proportion expressed as a ratio of the weights of its constituents is approximately:

Cement: Fine Aggregate: Coarse Aggregate

= 1: 1: 2

This exact mix proportion was meticulously maintained for all three batches of concrete (Jar Water, Tap Water, and Salty Water), ensuring that the type of water used was the sole variable under investigation.

3.3 Casting of Specimens

The casting of all concrete specimens was performed meticulously to ensure uniformity and compliance with standard testing protocols. The entire process, from mixing to finishing, was conducted for each of the three batches of concrete (Jar Water, Tap Water, and Salty Water).

3.3.1 Mould Preparation

Standard cast iron cube moulds of internal dimensions 150mm x 150mm x 150mm were used for casting the compressive strength test specimens. Prior to casting, the inner surfaces of the moulds were thoroughly cleaned to remove any old concrete or dust. To ensure easy de-moulding without damaging the specimens, a thin, even layer of mould-releasing oil was applied to all interior faces of the assembled moulds.

3.3.2 Mixing Process

The mixing was conducted on a non-porous, watertight platform. The required quantities of cement and fine aggregate were first mixed in a dry state until a uniform colour was achieved. The coarse aggregate was then added and mixed thoroughly with the cement-sand mixture. Finally, the pre-measured water (Jar, Tap, or Salty) was gradually added to the dry mix, and the entire mass was turned over repeatedly until a homogeneous concrete of consistent colour and workability was achieved.

3.3.3 Moulding and Compaction

The prepared concrete was placed into the oiled moulds in three equal layers, with each layer being approximately 50mm thick. Each layer was compacted using a standard 16mm diameter tamping rod. A total of 25 uniformly distributed blows were applied to each layer to ensure full compaction and the removal of entrapped air, as stipulated in IS 516:2018. The blows were delivered with sufficient force to penetrate the full depth of the layer and were distributed evenly across the entire surface area.

3.3.4 Finishing and Identification

After compacting the final layer, the surface of the concrete was finished level with the top edge of the mould using a trowel, creating a smooth, flat surface. Each specimen was immediately and carefully marked with a unique identification code indicating the type of water used (e.g., JW, TW, SW) and the intended age of testing (7D) to avoid any confusion during the curing and testing phases. The cast specimens were then left undisturbed in a vibration-free environment for 24 hours before de-moulding.

Fig: Casting of Concrete Moulds of 150mm*150mm*150mm

3.4 Curing

Curing is a critical post-casting process that ensures sufficient moisture is available for the continued hydration of cement, which is essential for proper strength development and enhanced durability. A standardized curing procedure was followed for all specimens to ensure consistent and comparable results.

3.4.1 De-moulding

Approximately 24 hours after casting, once the concrete had achieved sufficient initial hardness, the specimens were carefully de-moulded from their cast iron forms. The identification marks on each cube were checked for clarity.

3.4.2 Curing Method

The chosen method for this study was water curing by complete immersion, which is considered the most effective method for promoting hydration. Immediately following de-moulding, all specimens were fully submerged in a large, clean water curing tank.

3.4.3 Curing Conditions and Duration

The water in the curing tank was maintained at a temperature of 27 ± 2 °C, in accordance with the specifications of IS 516:2018. This controlled environment provides optimal conditions for the chemical reactions of hydration to proceed.

The specimens were cured for 7-Day period to evaluate the rate of strength gain:

7-Day Curing: A set of cubes from each water type (Jar, Tap, and Salty) was removed from the tank after 7 days of continuous curing to determine their early-age strength.

All specimens, regardless of the mixing water used, were cured together in the same tank. This ensured that the curing conditions were identical for all samples, thereby isolating the type of mixing water as the sole experimental variable affecting the final strength. Specimens were only removed from the water just prior to being tested.

3.5 Tests Conducted

To evaluate the effect of different water types on the properties of concrete, two standard tests were performed: one on the fresh concrete and one on the hardened concrete. All tests were conducted in accordance with the relevant Indian Standard (IS) codes of practice.

3.5.1 Workability Test (Slump Test)

The workability and consistency of the fresh concrete for each mix (Jar Water, Tap Water, and Salty Water) were determined using the slump test. This test was performed immediately after the mixing process was completed for each batch.

- Standard: The procedure was carried out as per IS 1199:1959 - Methods of Sampling and Analysis of Concrete.
- Procedure: A standard slump cone was filled with fresh concrete in three layers of equal volume. Each layer was tamped 25 times with a standard tamping rod. After the top layer was filled and leveled, the cone was lifted vertically, and the resulting subsidence (slump) of the concrete was measured in millimeters (mm).

3.5.2 Compressive Strength Test

The primary objective of this project was to determine the compressive strength of the hardened concrete. The test was performed on the 150mm cube specimens at 7 days and 28 days.

- Standard: The procedure was carried out as per IS 516 (Part 1/Sec 1): 2018 - Hardened Concrete — Methods of Test.
- Procedure: After the specified curing period (7 days), the cube specimens were removed from the curing tank and their surfaces were wiped dry. The dimensions of the specimen were measured, and it was placed centrally on the platen of a Compression Testing Machine (CTM). The load was applied axially at a constant rate of 140 kg/cm ^{2}/min (approximately 14 N/mm^2/min) until the specimen failed. The maximum load at which the cube failed was recorded. The compressive strength was then calculated using the formula:

Compressive Strength (N/mm^2) = Maximum Load (N) / Cross-sectional Area (mm^2)

3.5.3 Number of Specimens

To ensure reliability and to account for any potential experimental variability, three identical cube specimens were tested for each condition (i.e., for each water type at each testing age). The average of the three results was taken as the representative compressive strength for that condition. The complete testing scheme is summarized in Table 3.2.

Table 3.2: Specimen Testing Scheme

Water Type	Test Age	Number of Cubes Tested
Jar Water (JW)	7 Days	3
Tap Water (TW)	7 Days	3
Salty Water (SW)	7 Days	3
		Total Cubes = 9

3.6 Experimental Setup

The experimental work was conducted using standard laboratory equipment to ensure the accuracy and reliability of the test results. The primary apparatus used for this investigation is described below.

3.6.1 General Apparatus

Weighing Balance: A calibrated electronic weighing balance with a capacity of 50 kg and an accuracy of ±10 g was used for measuring the quantities of cement, fine aggregate, and coarse aggregate.

Measuring Cylinder: A 1-liter capacity glass measuring cylinder was used for the precise measurement of the different types of water.

Mixing Platform & Tools: A non-porous, watertight steel tray was used for hand-mixing the concrete. Standard tools, including trowels and shovels, were used to facilitate the mixing process.

3.6.2 Slump Test Apparatus

The setup for the slump test conformed to IS 1199 and consisted of:

Slump Cone: A metallic mould in the shape of a frustum of a cone with internal dimensions:

Bottom Diameter: 200 mm

Top Diameter: 100 mm

Height: 300 mm

Tamping Rod: A standard steel tamping rod with a diameter of 16 mm and a length of 600 mm, with a rounded, hemispherical tip.

Measuring Scale: A steel ruler was used to measure the final slump value.

3.6.3 Compressive Strength Testing Setup

The compressive strength of the hardened concrete cubes was determined using a calibrated Compression Testing Machine (CTM).

Machine Type: A hydraulically operated, digital Compression Testing Machine.

Platens: The machine was equipped with hardened steel bearing platens. The upper platen had a self-aligning ball-seating arrangement to ensure the uniform application of load onto the specimen.

Load Application and Display: The machine had a rate-of-loading control system, which was set to apply the load at the specified rate of 14 $N/mm^2/min$. A digital display provided real-time load readings with a least count of 0.1 kN, allowing for an accurate recording of the ultimate failure load.

Photographs of the key equipment and testing procedures are included in Appendix [A].

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the experimental results obtained from the laboratory tests conducted on the fresh and hardened concrete specimens. The data is systematically presented in tables and graphs to facilitate comparison. This is followed by a detailed discussion that interprets these results, correlates them with established concrete technology principles, and explains the observed effects of the different water types.

4.1 Workability Results (Slump Test)

The workability of the fresh concrete for each of the three mixes was assessed using the slump test, conducted as per IS 1199. The objective was to measure the consistency and ease of placing the concrete. The slump value for each mix is recorded in Table 4.1.

Table 4.1: Slump Test Results

Water Type	***Slump Value***
Jar Water (JW)	
Tap Water (TW)	
Salty Water (SW)	

- **Interpretation of Slump Results:**

4.2 Compressive Strength Results

The compressive strength of the 150mm concrete cubes was determined at 7 days and 28 days for each water type. For each condition, three cubes were tested, and the average value was taken as the representative strength. The results are summarized in Table 4.2.

Table 4.2: Compressive Strength Test Results

Water Type	**Age (Days)**	**Average Compressive Strength (N/mm² or MPa)**
Jar Water (JW)	7	
Tap Water (TW)	7	
Salty Water (SW)	7	

- **Interpretation of Compressive Strength Results:**

4.3 Discussion

The experimental results clearly demonstrate that while potable water (Jar and Tap) yields excellent and reliable concrete, the use of saline water has a significant detrimental effect on its properties.

- **Effect of Impurities on Cement Hydration and Strength:**

- **Performance of Jar Water vs. Tap Water:**

- **Correlation between Workability and Strength:**

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

This chapter summarizes the principal findings derived from the experimental investigation and provides recommendations for both construction industry practices and future research based on the outcomes of this study.

5.1 Conclusion

Based on the experimental investigation into the effect of jar water, tap water, and salty water on the properties of concrete, the following conclusions can be drawn:

- **Effect on Workability:** The type of water used had a measurable effect on the workability of fresh concrete. The mixes prepared with Jar Water and Tap Water yielded higher slump values, indicating good workability. The mix prepared with Salty Water showed a noticeable reduction in slump, suggesting that the presence of dissolved salts makes the concrete stiffer and potentially more difficult to place and compact.
- **Effect on 7-Day Compressive Strength:** The 7-day compressive strength results showed that all three concrete mixes performed adequately, achieving a significant portion of their expected design strength. Interestingly, the concrete made with Salty Water exhibited a marginally higher compressive strength at 7 days compared to the mixes made with Jar and Tap Water.
- **Overall Finding:** This study confirms that while potable water (both purified and tap) provides reliable and consistent concrete properties, the use of salty water alters both workability and early-age strength development. The higher 7-day strength of the salty water mix is indicative of the accelerating effect of chlorides on cement hydration, as documented in the literature. However, these 7-day results alone are insufficient to judge the overall quality and long-term performance of the concrete.

5.2 Recommendations

The findings of this project, though limited to 7-day results, lead to several important recommendations.

❖ **Recommendations for Construction Practice:**

- **Adherence to Standards:** It is strongly recommended that construction professionals strictly adhere to the guidelines of IS 456:2000, which advocate for the use of clean, potable water. The consistent performance of Tap Water in this study validates its use as a safe and reliable option.
- **Caution Against Salty Water:** Despite the acceptable or even higher 7-day strength, the use of salty water in reinforced concrete construction must be avoided. The literature review and established knowledge confirm that the short-term strength gain is overshadowed by severe long-term risks, including significantly lower ultimate (28-day) strength and a high probability of reinforcement corrosion.

- **Avoid Misinterpretation of Early Strength:** Site engineers should be cautious when interpreting early-age strength data, especially when non-standard materials are used. An accelerated 7-day strength gain, as seen with salty water, is not an indicator of superior quality but rather a chemical side-effect that often foretells poor long-term performance.

❖ Recommendations for Further Study:

- **Conduct 28-Day Strength Testing (Critical):** The most crucial next step is to extend this study by testing specimens at 28 days. This is essential to quantify the expected reduction in ultimate compressive strength caused by salty water and provide a complete and non-misleading conclusion. Testing at 56 or 90 days would provide even deeper insights into long-term strength development.
- **Investigate Durability Aspects:** Future research should focus on the durability of concrete made with these water types. Tests such as water permeability, sorptivity, and a half-cell potential test for reinforcement corrosion would provide a comprehensive understanding of the long-term service life risks.
- **Expand the Scope of Contaminants:** The study could be expanded to include other common water contaminants, such as sulfates, industrial effluents, or high organic content, to build a more extensive database on water quality effects.
- **Study Water Purification Methods:** Given the scarcity of fresh water in many regions, a valuable future project would be to investigate simple, cost-effective methods for treating saline or brackish water to make it suitable for use in concrete.

CHAPTER 6: REFERENCES / BIBLIOGRAPHY

The following books, research publications, and Indian Standard codes were referred to during the course of this project to establish the theoretical background and procedural guidelines.

❖ Books and Publications

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❖ Indian Standard (IS) Codes

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Note: The citation style used here is APA 7th Edition.

CHAPTER 7: APPENDICES

Fig: Compression Testing Machine (CTM)

Fig: Standard Slump Cone

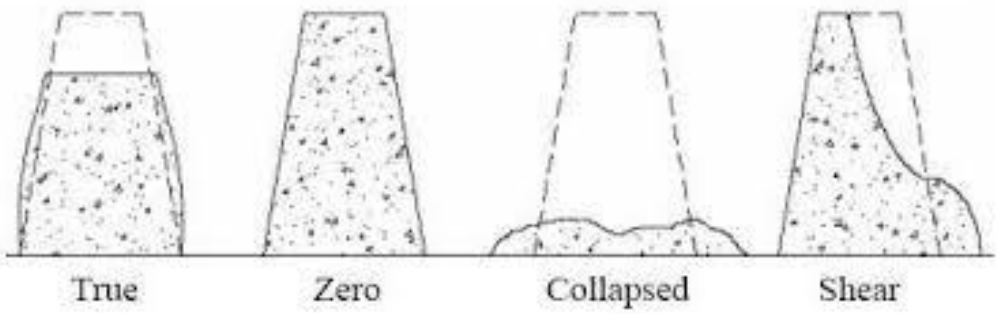

Fig: Types of Slump

Fig: Concrete Moulds